

Semantic Services

Inbound - Ingestion Pipeline

Microservices Architecture Each stage of the pipeline is implemented as a microservice to ensure modularity and scalability:

- **Extraction Service:** Retrieves data from GCP, including tables and XML files, ensuring proper access control and permissions.
- **Transformation Service:** Converts data into RDF format following established ontological conventions.
- **Validation Service:** Ensures RDF compliance with RDF 1.0 standards and schema constraints.
- **Loading Service:** Loads RDF data into Metaphactory and GraphDB through API endpoints.
- **Monitoring & Reporting Service:** Tracks pipeline status, error handling, and audit logs for governance.

Extraction from Google Cloud Platform (GCP)

- **Catalog Interface:** Develop a front-end application that interfaces with the existing catalog of assets stored in GCP. Users can select specific structures, such as tables or XML files, for processing.
- **Data Extraction:**
 - Extracts tables from BigQuery datasets.
 - Retrieves XML files from GCP Storage.
 - Ensures proper metadata tagging for downstream use.

Transformation to RDF

- **Standard Naming Conventions:** Establish a consistent naming convention for ontologies, including title, description, and IRI qualifiers.
- **Data Mapping and Conversion:**
 - Maps extracted structures to predefined ontologies.
 - Converts structured data to RDF triples using Turtle format.
 - Ensures semantic consistency across datasets.
- **Integration with Google Gemini:**
 - Sends extracted data to Gemini via API request.
 - Receives enriched output, including inferred relationships and entity classifications.
 - Parses Gemini's output to extract relevant RDF triples.
 - Formats the response into a Turtle file ensuring RDF 1.0 compliance.
 - Stores the Turtle file in a GCP staging area for validation and ingestion.

Staging and Validation

- **Staging Area:** Store the generated RDF files in a designated GCP staging area before final processing.

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- **Validation Mechanism:**
 - Parses RDF data to check for syntax errors.
 - Verifies compliance with RDF 1.0 and schema constraints.
 - Implements automated checks to prevent data corruption.

Loading into Metaphactory and GraphDB

- **Data Ingestion Process:**
 - Uses Metaphactory APIs to load RDF triples.
 - Ensures transactional consistency during data uploads.
 - Applies version control to maintain lineage and traceability.

Monitoring and Reporting

- **Logging:** Implements structured logging for pipeline status tracking.
- **Alert Mechanism:**
 - Generates notifications for validation errors.
 - Sends alerts for failures in data ingestion and processing.
 - Supports real-time monitoring dashboards.
- **Audit Trails:** Maintains an immutable log of all transformations and transactions.

User Interface for Structure Selection

- **Front-End Application:**
 - Provides an intuitive UI for browsing and selecting data structures from the GCP catalog.
 - Displays metadata, lineage, and processing status.
 - Offers guided workflows for onboarding new assets.